

Elementary Transfer Principles for Supercongruence Towers

A compact portfolio draft for external review

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Working draft. The arguments are complete proof candidates with exact transcription checks, but the manuscript has not been peer reviewed or submitted. The purpose of this version is focused mathematical review, especially by Peter Bala, Paul D. Hanna, and specialists in p -adic supercongruences.

Abstract

We collect four elementary mechanisms for adjacent prime-power supercongruences. First, a carry-and-transfer argument proves the all-prime congruence conjectured for A183068. Second, complete unit blocks prove a cubic transfer theorem for balanced products of gamma values at rational slopes. This theorem settles the congruence part of a packet of fractional-factorial conjectures and, together with known or finite-sum integrality, proves the full towers for A364175, every row of A365025, and every homogeneous row of A364513. Third, the same unit-block method gives uniform half-integral binomial families containing A091527, A262732, A275652, and A275654. Fourth, a complementary-factor decomposition of MacMahon's product proves a nonlinear p^{4r} Frobenius tower for plane partitions in every $N \times N \times cN$ box, including A008793, A352656, and A352657. We separate theorem, integrality, finite verification, and literature-priority status throughout. A final section packages the towers as arithmetic state sums and explains the precise, limited analogy with spin-foam coarse graining.

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1 Introduction and status boundary

An adjacent prime-power supercongruence compares the values of an integer sequence at two successive points of a p -adic tower. The typical form is

$$a(np^r) \equiv a(np^{r-1}) \pmod{p^{\lambda_r}},$$

while product sequences can obey the nonlinear variant

$$a(np^r) \equiv a(np^{r-1})^p \pmod{p^{\lambda_r}}.$$

The main theme here is that many apparently unrelated OEIS conjectures are controlled by one of two local operations:

1. discard summands whose base- p carries already supply the modulus, and rescale the remaining summands; or
2. split a product into its p -divisible factors and complete blocks of p -adic units, then annihilate the first two reciprocal moments.

The source conjectures were formulated by Peter Bala on records created by several OEIS contributors. In particular, Paul D. Hanna created A183068, and Bala supplied its factorial-sum formula and supercongruence conjecture. The present manuscript attributes the source statements but does not assign authorship to their originators without their consent.

There are three separate questions for every claimed application.

1. Is the displayed value an integer (or at least p -integral)?
2. Does the adjacent quotient have the asserted p -adic precision?
3. Is the result new in the literature?

The proofs below address the first two only where stated. Negative searches are not treated as priority certificates.

2 Unit blocks and classical binomial transfer

We write $v_p(x)$ for the normalized p -adic valuation of a nonzero rational number. A congruence between rational numbers is interpreted in $\mathbb{Q}_p$.

Lemma 2.1 (complete unit block). *Let $p \geq 5$, let $P = p^e$ with $e \geq 1$, and let I be any union of complete intervals of length P . Summing only over integers prime to p ,*

$$\sum_{j \in I, p \nmid j} \frac{1}{j} \equiv 0 \pmod{P^2}, \quad \sum_{j \in I, p \nmid j} \frac{1}{j^2} \equiv 0 \pmod{P}.$$

Proof. For a single reduced residue block, multiplication by a unit c with $c^2 \not\equiv 1 \pmod{p}$ permutes the units. Hence the reciprocal-square sum is congruent to c^{-2} times itself and is zero modulo P . Pairing u and $P - u$ gives

$$\frac{1}{u} + \frac{1}{P - u} \equiv -\frac{P}{u^2} \pmod{P^2},$$

so the reciprocal sum vanishes modulo P^2 . Translation by tP preserves the two precisions because

$$(tP + u)^{-1} \equiv u^{-1} - tPu^{-2} \pmod{P^2}, \quad (tP + u)^{-2} \equiv u^{-2} \pmod{P}.$$

Adding blocks proves the result. $\square$

We also use the classical Ljunggren–Jacobsthal–Kazandzidis congruence in the following adjacent form: if $p \geq 5$, $N = pM$, and $e = v_p(N)$, then

$$\frac{\binom{aN}{bN}}{\binom{aM}{bM}} \equiv 1 \pmod{p^{3e}} \quad (1)$$

whenever the binomial coefficients are defined. Refined versions and the small-prime losses are discussed in [3, 8, 11].

3 Balanced rational gamma ratios

3.1 A rational-binomial transfer

For positive integers a, b, q with $a > bq$, define

$$G_{a,b,q}(N) = \binom{aN/q}{bN} = \frac{\Gamma(aN/q + 1)}{\Gamma(bN + 1)\Gamma((a - bq)N/q + 1)}.$$

Theorem 3.1 (rational-binomial transfer). *Let $p \geq 5$, $p \nmid q$, and $N = pM$. If $e = v_p(N)$, then*

$$\frac{G_{a,b,q}(N)}{G_{a,b,q}(M)} \equiv 1 \pmod{p^{3e}}.$$

Proof. The finite product

$$G_{a,b,q}(N) = \frac{1}{q^{bN}(bN)!} \prod_{j=0}^{bN-1} (aN - qj)$$

has a numerator factor divisible by p exactly when $p \mid j$. Cancelling those factors and the corresponding denominator factors against the level M product gives the exact identity

$$\frac{G_{a,b,q}(N)}{G_{a,b,q}(M)} = \prod_{\substack{1 \leq j < bN \\ p \nmid j}} \left(1 - \frac{aN}{qj}\right).$$

The sign disappears because $b(N - M) = bM(p - 1)$ is even. Put $P = p^e$. The index interval is a union of complete P -blocks. Lemma 2.1 shows that the linear and quadratic elementary-symmetric terms are divisible by P^3 . Every term of degree at least three contains N^3 , while all denominators are p -adic units. Thus the product is 1 modulo P^3 . $\square$

3.2 The residue-balanced theorem

Let $E : \mathbb{Q}_{>0} \rightarrow \mathbb{Z}$ have finite support and put

$$F(N) = \prod_{\alpha \in \mathbb{Q}_{>0}} \Gamma(\alpha N + 1)^{E(\alpha)}.$$

Assume

$$\sum_{\alpha} \alpha E(\alpha) = 0 \quad (2)$$

and, for every nonzero class $\rho \in \mathbb{Q}/\mathbb{Z}$,

$$\sum_{\alpha \equiv \rho \pmod{\mathbb{Z}}} E(\alpha) = 0. \quad (3)$$

Let Q be a common denominator of the slopes.

Theorem 3.2 (residue-balanced gamma transfer). *For $p \geq 5$, $p \nmid Q$, and $n, r \geq 1$,*

$$\frac{F(np^r)}{F(np^{r-1})} \equiv 1 \pmod{p^{3r}}.$$

If $F(np^{r-1})$ is p -integral, then

$$F(np^r) \equiv F(np^{r-1}) \pmod{p^{3r}}.$$

Proof. Condition (3) pairs every numerator gamma factor of nonintegral slope with a denominator factor in the same residue class. For $\alpha > \beta$ and $d = \alpha - \beta \in \mathbb{Z}_{>0}$,

$$\frac{\Gamma(\alpha N + 1)}{\Gamma(\beta N + 1)} = \binom{\alpha N}{d N} \Gamma(d N + 1).$$

Use the inverse identity if $\alpha < \beta$. Repeating this operation leaves rational binomials governed by Theorem 3.1 and a balanced integer-slope factorial ratio. A balanced integer-slope ratio factors as a Laurent product of $\binom{jN}{N}$, so (1) applies to every factor. Products and inverses preserve the congruence. Multiplication by the lower p -integral value proves the final assertion. $\square$

3.3 Applications

The sequence A364175 is

$$A(N) = \frac{(6N)!(2N/3)!}{(3N)!(2N)!(5N/3)!} = \binom{6N}{3N} \binom{3N}{N} \binom{5N/3}{N}^{-1}.$$

David Radcliffe proved its integrality [5]. Theorem 3.2 therefore gives the following complete tower.

Corollary 3.3. *For every prime $p \geq 5$ and $n, r \geq 1$,*

$$A(np^r) \equiv A(np^{r-1}) \pmod{p^{3r}}.$$

For a fixed $s \geq 0$, define

$$U_s(N) = \frac{\Gamma(N/2 + 1) \Gamma((2s + 1)N + 1) \Gamma((2s + 1/2)N + 1)}{\Gamma(N + 1) \Gamma(sN + 1)^2 \Gamma((s + 1/2)N + 1)^2}.$$

The OEIS identity

$$U_s(N) = \sum_{j=0}^{sN} \binom{(2s + 1)N}{sN - j}^2 \binom{N + j - 1}{j}$$

proves integrality. The residue class $1/2 + \mathbb{Z}$ has two numerator and two denominator factors, so Theorem 3.2 proves the conjectured cubic tower for every row of A365025, including A365026 and A365027.

For $s \geq 3$, the row family on A364513 has the gamma form

$$V_s(N) = \frac{s-2}{s} \frac{\Gamma((s+2)N+1)\Gamma(sN/2+1)^2}{\Gamma((s+2)N/2+1)\Gamma(sN+1)\Gamma((s-2)N/2+1)\Gamma(N+1)^2}$$

and the integral finite sum

$$V_s(N) = \sum_{j=0}^N \binom{sN-1}{N-j}^2 \binom{(s-2)N+j-2}{j}.$$

The constant $(s-2)/s$ cancels from adjacent quotients. Theorem 3.2 proves the tower for every fixed $s \geq 3$, including A364515, A364516, and A364517. Row 1 is not homogeneous and is not claimed here.

The same residue test applies to the displayed gamma quotients on A364172–A364184. It proves their cubic transfer in $\mathbb{Q}_p$; an integer congruence still requires global integrality. This logical separation is essential.

4 The all-prime A183068 theorem

Define

$$a(N) = \sum_{k=0}^N F(N, k), \quad F(N, k) = \frac{(2N+2k)!}{k!^4(N-k)!^2}.$$

The summand is the six-part multinomial coefficient

$$F(N, k) = \binom{2N+2k}{k, k, k, k, N-k, N-k}.$$

Lemma 4.1 (carry depth). *Suppose $p^t \mid N$ and $s = v_p(k) < t$. Then*

$$v_p(F(N, k)) \geq 2(t-s).$$

For $p = 2$ the lower bound improves to $2(t-s) + 1$.

Proof. At level $q = p^i$, write $N = qM$ and $k = qa + u$ with $0 < u < q$. Legendre's formula gives the level contribution

$$\left\lfloor \frac{2N+2k}{q} \right\rfloor - 4 \left\lfloor \frac{k}{q} \right\rfloor - 2 \left\lfloor \frac{N-k}{q} \right\rfloor = 2 + \left\lfloor \frac{2u}{q} \right\rfloor \geq 2.$$

There are $t-s$ active levels. If $p = 2$, then at $i = s+1$ one has $u = 2^s$, and this level contributes 3. □

We use the multinomial form of the prime-power Ljunggren–Jacobsthal–Kazandzidis estimate. If every positive component b_i is divisible by p^s , then

$$\frac{\binom{p \sum b_i}{pb_1, \dots, pb_m}}{\binom{\sum b_i}{b_1, \dots, b_m}} \equiv 1 \pmod{p^{3(s+1)-\epsilon_p}}, \quad (4)$$

where $\epsilon_2 = 2$, $\epsilon_3 = 1$, and $\epsilon_p = 0$ for $p \geq 5$. The possible binary sign is harmless here: for $s \geq 1$ its exceptional parity condition cannot occur, while at $s = 0$ the modulus is 2.

Lemma 4.2 (summand transfer). *Let $N' = np^{r-1}$ and $0 \leq \ell \leq N'$. Then*

$$F(pN', p\ell) \equiv F(N', \ell) \pmod{p^{2r}}.$$

Proof. Apply (4) to

$$\ell, \ell, \ell, \ell, N' - \ell, N' - \ell.$$

Let s be the smallest valuation among positive components. If $s < r - 1$, Lemma 4.1 contributes $2(r - 1 - s)$ powers (one more at $p = 2$), while (4) contributes $3(s + 1) - \epsilon_p$. Their sum is at least $2r$. If $s \geq r - 1$, the scaling estimate alone suffices except possibly for $p = 2, r = 1$. In that case some positive component b is repeated, so the multinomial contains the even factor $\binom{2b}{b}$. This supplies the missing binary power. $\square$

Theorem 4.3 (A183068). *For every prime p and all $n, r \geq 1$,*

$$a(np^r) \equiv a(np^{r-1}) \pmod{p^{2r}}.$$

Proof. Set $N = np^r$. If $p \nmid k$, Lemma 4.1 gives $F(N, k) \equiv 0 \pmod{p^{2r}}$. The indices divisible by p are $k = p\ell$, and Lemma 4.2 replaces their summands by $F(N/p, \ell)$. Summing gives exactly $a(N/p)$ modulo p^{2r} . $\square$

5 Half-integral coefficient families

For $m \geq 2$, define

$$A_m(N) = [x^N] \left(\frac{(1+x)^m}{(1-x)^{m-2}} \right)^N.$$

Residue extraction gives

$$A_m(N) = 4^N \binom{(mN-1)/2}{N} = \frac{2^N}{N!} \prod_{j=0}^{N-1} (mN - (2j+1)). \quad (5)$$

Theorem 5.1 (odd-unit block family). *For $m \geq 2$, $p \geq 5$, and $n, r \geq 1$,*

$$A_m(np^r) \equiv A_m(np^{r-1}) \pmod{p^{3r}}.$$

Proof. Let $N = np^r$ and $M = N/p$. In (5), the factors indexed by odd multiples of p reproduce the level M product. The remaining quotient factors as

$$\frac{\binom{2N}{N}}{\binom{2M}{M}} \prod_{\substack{1 \leq t < 2N \\ t \text{ modd}, p \nmid t}} \left(1 - \frac{mN}{t} \right).$$

The first factor is 1 modulo p^{3r} by (1). The odd indices form complete unit blocks modulo $p^{v_p(N)}$; their first two reciprocal moments vanish to the precisions of Lemma 2.1. Expanding the finite product proves the same cubic precision for the second factor. Integrality follows from the coefficient definition. $\square$

The cases $m = 3$ and $m = 5$ prove the conjectures on A091527 and A262732. Both fail at $p = 3$, so the prime restriction is sharp for these named cases.

For $a \geq 3$, set

$$D_a(N) = \sum_{k=0}^N \binom{(a-1)N - k - 1}{N - k} \binom{aN}{k}^2.$$

Dixon's terminating ${}_3F_2(1)$ evaluation yields

$$D_a(N) = \binom{aN}{2N} \binom{(a+2)N/2}{N} \binom{2N}{N} \binom{aN/2}{N}^{-1}.$$

Theorem 3.1 with $q = 2$, together with (1), proves

$$D_a(np^r) \equiv D_a(np^{r-1}) \pmod{p^{3r}} \quad (p \geq 5).$$

The cases $a = 3, 5$ are A275652 and A275654; row $a = 7$ is A364304. The finite sum supplies integrality.

6 Symmetric-box plane partitions

For $c, N \geq 1$, let

$$B_c(N) = \prod_{i,j=1}^N \frac{cN + i + j - 1}{i + j - 1}.$$

MacMahon's formula identifies this with the number of plane partitions in an $N \times N \times cN$ box.

Theorem 6.1 (symmetric-box Frobenius tower). *For every prime p and all $c, n, r \geq 1$,*

$$B_c(np^r) \equiv B_c(np^{r-1})^p \pmod{p^{4r}}.$$

Proof. Pair the multiplicity- s and multiplicity- $(2N - s)$ factors in MacMahon's product to obtain

$$B_c(N) = (c+1)^N \prod_{s=1}^{N-1} \left(1 + \frac{c(c+2)N^2}{s(2N-s)} \right)^s.$$

After replacing N by pN , the factors with $p \mid s$ reproduce the p -th power of the level N product. The residual unit product is

$$U_{p,c}(N) = \prod_{\substack{1 \leq s < pN \\ p \nmid s}} \left(1 + \frac{c(c+2)p^2N^2}{s(2pN-s)} \right)^s.$$

For odd p , the logarithmic linear term is a multiple of the reciprocal sum over the units in the interval $(np^r, 2np^r)$. Pairing t with $3np^r - t$ shows that this reciprocal sum vanishes modulo p^{2r} . The linear term therefore has valuation at least $4r$, and every higher logarithmic term has at least the same valuation.

For $p = 2$, complementary odd residues give the required reciprocal-block bound. If c is even, $c(c+2)$ supplies three bonus powers. If c is odd, the residual product is initially two powers short, but $B_c(N)$ is even: for odd N , box complementation is fixed-point-free, and Legendre's formula gives $v_2(B_c(2N)) = 2v_2(B_c(N))$. Thus the square $B_c(N)^2$ supplies the two missing powers. $\square$

The cases $c = 1, 2, 3$ prove the named conjectures on A008793, A352656, and A352657.

7 A spin-foam-inspired state-sum interpretation

Spin-foam models assign local labels to the faces and edges of a two-complex, multiply local amplitudes, and sum over compatible labelings [2, 6]. Their renormalization problem compares amplitudes on fine and coarse discretizations [10]. The supercongruence towers above admit a precise arithmetic analogue of this *state-sum and coarse-graining architecture*, although they are not models of quantum gravity.

For A183068, a state at scale N is a six-color occupancy vector

$$(k, k, k, k, N - k, N - k)$$

with local multiplicity $F(N, k)$. The partition function is $a(N) = \sum_k F(N, k)$. The coarse-graining map at the prime p sends the surviving state $k = p\ell$ to ℓ . States not lying in the image acquire at least $2r$ powers of p from carries, while the amplitude of a surviving state agrees with its coarse amplitude modulo p^{2r} . Thus Theorem 4.3 is a local-to-global invariance statement for an arithmetic partition function.

The product models have the same shape with no external summation. A microscopic label is an individual numerator or denominator factor. The p -divisible labels reproduce the coarser product exactly, and the unit labels form complete fibers over $(\mathbb{Z}/p^e\mathbb{Z})^\times$. Lemma 2.1 says that the first two moments of every fiber vanish to the needed p -adic precision. In this language the passage

$$N \mapsto pN$$

is an arithmetic subdivision, and a congruence such as

$$Z(pN) \equiv Z(N) \pmod{p^{\lambda r}}$$

or $Z(pN) \equiv Z(N)^p$ is a finite-precision renormalization fixed-point law. The exponent λr measures the depth to which fine-scale fluctuations are invisible in the coarse observable.

This viewpoint suggests a reusable definition. An *arithmetic foam* is a finite labeled cell system with amplitudes in $\mathbb{Z}_p$, together with a surjective coarse-graining map whose nontrivial fibers have controlled p -adic moments. A state-sum supercongruence follows when

1. amplitudes outside the coarse image vanish to the target depth;
2. amplitudes inside the image transfer to their coarse counterparts; and
3. these estimates are stable under summation or multiplication.

The carry lemma and the unit-block lemma are two concrete realizations of the same three axioms.

The analogy stops there. No gauge group, representation labels, intertwiner spaces, Pachner-move invariance, semiclassical limit, or continuum theory has been constructed. Consequently these arithmetic foams should be read as a physics-inspired organizational device, not as a claim that the OEIS sequences are spin-foam quantum-gravity amplitudes. The useful research question is narrower: whether known spin-foam and tensor-network coarse-graining identities suggest new finite-fiber moment cancellations, and whether those cancellations can then be proved purely arithmetically.

8 Compact portfolio ledger

Table 1 records the named OEIS consequences presently packaged as complete proofs or explicit deductions in the public repository. It is a review index, not a claim that every line uses the same proof. The four engines proved in this manuscript cover the rows marked “this paper.” The remaining rows have self-contained companion notes and exact checkers in the repository and should be audited separately before being incorporated into a final arXiv submission.

Table 1: Named supercongruence consequences in the review portfolio.

OEIS records	Mechanism	Review status
A183068	multinomial carries and scaling	Theorem 4.3; all primes
A008793, A352656, A352657	symmetric-box product	Theorem 6.1; all primes
A364175, A365025–A365027, A364515–A364517	residue-balanced rational gamma ratios	Theorem 3.2; $p \geq 5$
A091527, A262732	odd unit blocks	Theorem 5.1; $p \geq 5$
A275652, A275654, A364304	Dixon and half-binomial transfer	Section 5; $p \geq 5$
A061164, A364506	balanced factorial or binomial Laurent products	complete deductions from (1)
A288470, A141057	termwise carry/Frobenius families	complete companion proofs; external review pending
A357509, A357568	binomial-quotient cancellation	complete companion proofs; external review pending
A002003, A348410, A351857, A352373, A370101, A370102	coefficient framing/Cartier transfer	complete companion proofs; external review pending
A008485, A008705, A270913, A270919	modular-product prime coefficients	complete companion proofs; external review pending
A255672, A270922, A270924	colored Euler-product transfer	complete companion proofs; external review pending
A108625, A143007, A177316	multivariate Apéry transfer at the boundary prime	complete companion proofs; external review pending
A049505	paired symmetric-plane-partition products	complete companion proof; external review pending
A357510	odd Apéry moment at primes	complete companion proof; composite extensions remain separate

The repository also contains reductions and computational targets. They are intentionally absent from Table 1; in particular, the enhanced Apéry defect relations, paired ordinary/shifted towers, and the remaining global integrality problems are not declared solved.

9 Verification and review questions

The exact checkers use integer or rational arithmetic and are intended to catch transcription errors, not to replace proofs. The main commands are

```
python verification/verify_a183068.py
```

```
python verification/related/verify_rational_gamma_ratio_towers.py
python verification/related/verify_symmetric_box_plane_partitions.py
python verification/related/verify_dixon_legendre_towers.py
python verification/check_repository_integrity.py
```

The rational-gamma checker performs 8,666 exact checks; the A183068, symmetric-box, Dixon–Legendre, and repository-integrity suites are also registered by the public repository runner.

For an efficient independent review, the highest-risk obligations are:

1. verify the exact cancellation identity in Theorem 3.1;
2. verify that residue-class balance is sufficient for the pairing in Theorem 3.2, including negative multiplicities;
3. audit the small-prime exponent and sign in (4), especially $p = 2, r = 1$;
4. audit the binary parity repair in Theorem 6.1; and
5. check each OEIS formula and the literature-priority boundary independently of the proofs.

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